

or 24V DC supply can be used to power the motors, either via an AC-DC adaptor or a suitable battery. The Arduino must be powered separately with a stable 5V regulated supply, such as through a USB connection or an external 5V source, to ensure reliable performance.

All required electronic components are listed in Table 1 for the Bill of Materials, while the mechanical parts, materials, and tools are listed in Table 2.

## Circuit diagram

Fig. 2 shows the circuit diagram of the full-scale humanoid robotic arm. The system is built around an Arduino Uno (MOD1) and a PCA9685 servo driver module (MOD2), along with multiple servo motors, including high-torque (25kg·cm), heavy-duty (DH03-X/ASME-MRB), and micro servos (SG90/MG90S), supported by essential electrical components.

The diagram illustrates how multiple servos are controlled efficiently using the PCA9685 module. The Arduino communicates with the PCA9685 via the I<sup>2</sup>C interface through the SDA and SCL pins, minimising the number of GPIO pins required.

The PCA9685 provides up to 16 PWM output channels, each connected to a servo motor (S1-S11 shown). The signal (SIG) lines are driven by the PCA9685, while all servos share common power (V+) and ground (GND). A dedicated 12V supply (regulated as required) powers the servos to ensure adequate current for high-torque operation. It is essential to maintain a common ground between the Arduino, PCA9685, and the power supply for stable performance.

This design uses an Arduino Uno with a PCA9685 driver. Connect each servo to the appropriate channel based on its function (fingers, wrist, elbow, and shoulder). If using an Arduino Mega, servos can also be driven directly from PWM pins, but the

| Category                    | Item Description           | Quantity | Specifications/Notes                |
|-----------------------------|----------------------------|----------|-------------------------------------|
| Mechanical hardware         | M3 screws (8mm)            | 50       | Standard                            |
|                             | M3 screws (12mm)           | 15       | Standard                            |
|                             | M3 screws (18mm)           | 10       | Standard                            |
|                             | M3 screws (20mm)           | 14       | Standard                            |
|                             | M3 nuts                    | 20       | Standard                            |
|                             | M4 screws (8mm)            | 4        | Standard                            |
|                             | M4 screws (20mm)           | 10       | Standard                            |
|                             | M4 nuts                    | 10       | Standard                            |
|                             | M3 washers                 | 5        | Standard                            |
|                             | M4 washers                 | 3        | Standard                            |
|                             | General screws and nuts    | —        | Miscellaneous                       |
| Actuation accessories       | Servo arm (25T)            | 1        | Usually included with servo         |
| Wiring and electrical       | Insulating tape            | 1 roll   | —                                   |
|                             | Zip ties                   | ~50      | 10cm                                |
| 3D printing materials       | PLA filament               | 2kg      | 1.75mm                              |
|                             | Nylon filament (optional)  | 200g     | For high-stress parts               |
| Additional mechanical parts | Fishing nylon thread       | 1 roll   | 0.4mm (finger actuation)            |
|                             | Aluminium servo bracket    | 1        | ~58mm×37mm×25.5mm                   |
| Tools required              | Wire stripper              | 1        | Self-adjusting                      |
|                             | Standard pliers            | 1        | —                                   |
|                             | Cutting pliers             | 1        | —                                   |
|                             | Screwdriver set (Phillips) | 1 set    | —                                   |
|                             | Hex screwdriver            | 1        | —                                   |
|                             | Soldering iron             | 1        | —                                   |
|                             | Solder wire                | 1 roll   | Tin                                 |
|                             | Multimeter                 | 1        | —                                   |
|                             | Drill machine              | 1        | —                                   |
|                             | Drill bit set              | 1        | 2.78mm commonly used                |
|                             | Bench vise                 | 1        | —                                   |
|                             | 3D printer                 | 1        | Minimum 15cm×15cm×25cm build volume |

PCA9685 approach is recommended for reliable multi-servo control.

## Software

Since standard servo motors are used, the programming remains fairly simple, though it can become more advanced depending on user requirements and experience.

If using an Arduino with more than eight PWM outputs, the standard Servo library (Servo.h) can be used. However, when using the PCA9685 expansion module, the HCPCA9685.h library is required instead.

Begin by assigning an initial position to each servo and defining its minimum and maximum movement limits. Once configured, the robotic

arm can be programmed to perform the desired motions. For safety and accuracy, it is advisable to program and test each servo individually until the expected behaviour is achieved. Afterwards, all movements can be integrated into a single program.

In this system, the PCA9685 expansion module is used, and the code enables simultaneous opening and closing of all fingers. For elbow movement, a for loop is implemented to move the servo from 60° (fully down) to 200° (fully up) and back again. Adjusting the delay within the loop allows control over movement speed. Fig. 3 shows the code snippet for setting the servo joint pins and configuring the movement limits for each joint.